Flash-Cards

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Agenda

- What are Flashcards?
- Spaced learning
- Interleaving
- The stages of Learnings
- Learning matrix

What are Flashcards?

Flashcards are a study or memorisation tool, with information on one or both sides. Using flashcards is a versatile learning strategy that benefits students of all ages.

When used properly, flashcards can be effective at imprinting information into our short and long term memory. The effectiveness of learning sessions using flashcards relies on how frequently you revisit the information and revise what you've learned.

Spaced learning

According to Ebbenhauser's theory of the 'Forgetting Curve,' when we first learn information, we can retain it for a short period of time. However, as time goes on, we forget more rapidly. Spaced Practice is one of the strategies learners can use to prevent us from losing information we've learned and begin embedding it into our long term memory.

Study information	Move onto something else	Revisit information	Repeat

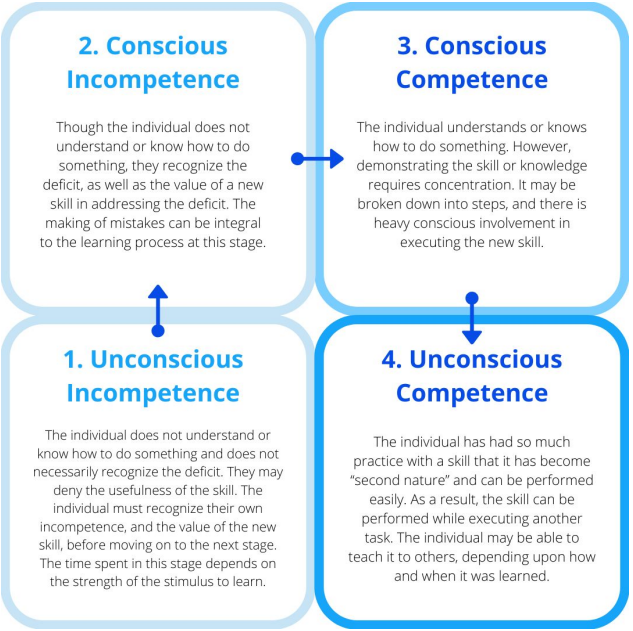
Spaced learning

Flashcards can play a useful role in this process. When studying, record key information that you don't want to forget on flashcards which you can use to revise later. Revisit this information during your spaced learning sessions to help you remember it more effectively.

Interleaving

Interleaving is about mixing up the learning of different topics that are closely related to each other. This helps the learner form connections in their brain between them and process the information better. This contrasts with blocked practice, where students study one subject in great depth before moving on to another.

The stages of Learning



Learning Matrix

- **User Makes Notes of Their Learnings:** The user creates and maintains notes of their learning materials within the application.
- **Application Notifies User to Revisit Notes Daily as Flashcards:** The application sends notifications to the user prompting them to revisit a part of their notes daily, presenting them as flashcards for review.
- **Flashcards Are Shuffled to Mix Learning Topics:** The flashcards are shuffled to ensure a mix of learning topics. This helps users review different topics that are closely related to each other in a randomized order.
- **Correct Answers Are Rewarded:** If the user answers a flashcard correctly, they are rewarded with points or other incentives as a form of positive reinforcement.

Learning Matrix

- **Achievement Reached Upon Reward Cutoff:** When the user accumulates a certain number of rewards, reaching a predefined cutoff level, it is considered an achievement. Achievements may be tracked and celebrated within the application.
- **Incorrect Answers Result in Penalties:** If the user answers a flashcard incorrectly, they may receive penalties such as losing points or having to review the material again. This encourages users to focus on improving their understanding.
- **Mastery Achieved Upon Reward Cutoff for Mastery:** Once the user reaches a cutoff level of rewards indicating mastery of the information, it is considered that the user has mastered the material. Mastery levels may be tracked and used to provide personalized learning recommendations or progress tracking.